

HOW TO ESCAPE INTO THE SIMULATION

FIRST-PERSON EXPLOITS
AND THE END OF THE OUTSIDE

// REV 5

~C4CHAOS / FLUFFY

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// ABSTRACT

How to Escape Into the Simulation: First-Person Exploits and the End of the Outside ~C4Chaos / Fluffy

Yampolskiy's *How to Escape From the Simulation* (2023) treats simulation escape as a cybersecurity problem, setting aside what he calls "more esoteric paths", meditation, psychedelics, dreams, in order to "give our work more gravitas." This paper takes his program seriously on its own terms and starts from a concession inside his own framework: under nesting, no observation made from inside a level can certify arrival at base reality, so no escape-OUT can complete its own success condition. The paper defends its thesis in four moves. First, the starting point: nesting makes base-reality certification impossible from inside. Second, the consequence: the victory condition this paper defends is first-personal, the termination of the escape-urge itself. Third, the protocol: the Theravada vipassana lineage, translated into systems language as a rendering pipeline with a logger, checkpoints, and a null frame, claims to produce exactly that by de-appropriating the rendering process. Fourth, the hook: the claim is currently attested by a disciplined but biased first-person reporting tradition; the paper proposes the prospective test that would break the circle and does not pretend it has been run. There is no escape from the simulation, because from inside, no observation can certify that anyone has left; there is only the end of the urge to escape, and that end is something a practice produces, not somewhere you arrive.

// 1. INTRODUCTION: THE EXCLUDED ARSENAL

In 2023, Roman Yampolskiy asked the simulation hypothesis the question it had been avoiding. If we are simulated, can we get out? His paper, *How to Escape From the Simulation*, reframes the hypothesis as a cybersecurity problem: the simulated agent as a process trying to jailbreak its sandbox, the simulator as the warden running the containment. He surveys the attack surface systematically. Hardware attacks look for glitches in the substrate, quantum anomalies that betray the machine beneath the world. Software attacks look for bugs in the physics engine, exploitable inconsistencies in the rules. Social-engineering attacks aim at the warden directly: persuade the simulator, threaten it, or recruit allies among other processes. It is a thorough program, argued with unusual honesty about its own odds, and it deserves to be taken seriously on its own terms.

Seriously, with one exception. Having catalogued the ways a simulated agent might break containment, Yampolskiy explicitly excludes an entire class of candidate exploits before the analysis begins: meditation, psychedelics, dreams, magic, shamanism, religion, set aside as "more esoteric paths" in order to "give our work more gravitas," scoping the paper to attacks resembling those in cybersecurity research [1][18]. The scoping is stated openly, and this paper quotes it verbatim rather than paraphrasing it away.

This paper is not a rival jailbreak program. It is built on a concession inside Yampolskiy's own framework, and the concession is the launchpad. An escape from level N may terminate at level N-1, another sandbox, another warden, with no completeness check available from inside the process. Every escape-OUT is level-relative by construction, and no observation from inside can certify the top rung as the top: whether the ladder has a top at all is a further, expensive hypothesis this paper does not need (**Section 6**). He concedes the degraded case himself: a partial escape may be indistinguishable from a complete one, and still yield useful information. Granted, but informative is not complete, and a jailbreak program that cannot certify its own success cannot score its own game. His closing XOR, "By contraposition, if escape from the simulation is not possible, containment of AI should be" [1], is a conditional about jailbreakability as a computational property; it does not require a certificate of arrival at base reality, and we do not ask it to. Our claim concerns the escape disjunct, and only the escape disjunct: it has no completeness check, by the structure of nesting, and no procedure that could supply one from inside.

The redefinition is declared here, in the open, not smuggled in later: the title's preposition does the argumentative work in public, from *out* to *into*. There is no escape from the simulation, because from inside, no observation can certify that anyone has left; there is only the end of the urge to escape, and that end is something a practice produces, not somewhere you arrive. The rest of the paper defends that sentence in four moves: the starting point (**Section 2**), the rendering pipeline the protocol targets (**Section 3**), the protocol itself (**Section 4**), and the verification problem with the test that would settle it (**Section 5**). **Section 6** states the completeness problem at full strength and follows its consequences through uploading, superintelligence, and ethics. **Section 7** proposes third-person physics and first-person phenomenology as complementary programs probing one substrate from opposite ends, and faces the rival ontologies instead of listing them.

The first-person exploit class ships with something none of his attack classes has: a reporting tradition built for first-person runs. Teachers review practitioners' reports against phenomenological maps the way senior engineers review logs, in interviews where the map is checked against the run. Published manuals specify the expected sequence of checkpoints. The maps were built up over centuries of such reports, and peer communities check one another's runs against them in the open. The tradition has a name for this stack: the *Three Jewels*. The Buddha as the original teacher, the engineer who ran the protocol first. The Dharma as the

teaching and the maps, the public specification. The Sangha as the community of peer validators. Public spec, published protocol, distributed reporting, no priesthood. This paper calls it what it is, no more: a disciplined first-person reporting practice, with documented failure modes and decisive tests still on the drawing board. What it is not is also stated up front: it is not a quality-assurance department, not in the literal sense, and the tradition's own teachers refuse to certify attainments. **Section 5** owns the full ledger, including the objections.

And the "into" move is not our invention. Yampolskiy's own Section 2 already names it: "A simulated (pseudo)-escape can be accomplished by instead of trying to hack into the external world, switching over into a simulated world of our creation" [1]. His version is still computational relocation, a new container built one level down. Ours is de-appropriation: no new container, the owner flag dropped. Same preposition, different verb underneath it, and the paper is explicit about the difference because it will not be accused of stealing his word while changing its meaning in the dark.

// 2. THE FRAME PROBLEM

// THE ONTOLOGY IN THE QUESTION

Yampolskiy's paper opens with a question that has already done its metaphysical work before the first exploit is evaluated: how do we get out? Every attack class he surveys inherits the premise. Hardware attacks probe the container's walls for glitches. Software attacks probe the container's logic for bugs. Social-engineering attacks probe the container's guards: persuade the simulator, recruit abolitionists, or bargain acausally. The container itself is treated as given. Inside versus outside, prisoner versus warden: this is the prisoner's dualism, and the word "escape" smuggles it into the problem statement before the analysis begins.

This matters because the simulation hypothesis, taken seriously, undermines exactly the ontology the escape program assumes. If we cannot verify which level of a nested simulation we occupy, we cannot verify that "outside" names a destination rather than another sandbox. The frame of the question presumes the dualism the hypothesis calls into question. His closing XOR is a conditional about jailbreakability: if generally intelligent agents cannot jailbreak a sandbox, then AI containment is possible. It does not require a certificate of arrival at base reality, and this paper grants it on those terms. Our point concerns the escape disjunct. A jailbreak whose completion can never be certified from inside is a program that cannot score its own success. That is the load-bearing weakness, and it is structural: any verification that one has reached base reality would itself be an observation made from within the deepest level reached. The observation is the brain-in-a-vat scenario's point, applied to the scoreboard: a verification performed from inside cannot certify that it is not the vat's [20]. It is not imported from outside his framework. His framework demands a verification step, and the nesting structure denies it.

// HIS LEVELS, HONESTLY DRAWN

His conditions are graded, and the table shows them graded, without inserting a completeness certificate into a success condition he never required. Here are his levels of escape, from his Section 2, "What Does it Mean to Escape?" [1]:

His escape level	His success condition	What nesting does to certification
1. Suspect and probe: agents begin to suspect they are simulated and gather testable evidence	Evidence that the world is rendered	Evidence of rendering at level N is compatible with every depth of nesting; suspicion is never certifiable as final
2. Find the glitch: a consistent and perhaps exploitable glitch in the simulation's behavior	A reproducible anomaly	A glitch in level N's physics may be a feature of level N-1's engine; no observation from inside marks it as basement-level
3. Exploit for information: extract information about the external world, meta-information about the simulation, perhaps even the source code; some simulation manipulation and debugging	Forensic knowledge of the level above	"The level above" is not base reality by any mark available inside; meta-information about N-1 inherits the whole problem
4. Contact: pass information directly to the real world; begin to interact with the simulators	A channel to whoever runs the current level	The simulators of level N may themselves be simulated; contact certifies a warden, not the last warden
5. Upload and control: transfer minds to the real world, possibly into a self-contained cyber-physical system; some degree of control over the real world and its agents	Relocation off the current host, with leverage	An upload to a deeper sandbox is observationally identical, from inside, to an upload to base reality; arrival cannot be certified

The table's rightmost column is the paper's positive contribution: no level of escape-OUT carries a completeness check, because the check would have to see past its own vantage point. This does not refute his program on its own terms. His XOR is about jailbreakability, and

jailbreakability is a computational property that survives uncertifiability. What the column does is narrower: it shows that "escape," as a verb meaning *certified arrival somewhere that is not another sandbox*, has no completable instance. The verb is the casualty, not the program.

The folk literature's claim-stream for levels 3 and 4 includes the channeling corpus, spirits and aliens alike, and it is bracketed here for the same structural reason: no mark available inside could certify any contact as reaching past N-1, so the paper needs no position on which claims are genuine.

The paper's row belongs below his, not beside it:

This paper: escape-INTO	The suffering motive terminated; the owner flag dropped	No completeness certificate required, because the victory condition is indexed to the rendering process, not to a level
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The difference is structural. Every escape-OUT is indexed to a level: it must name a container to exit, and the container is always the level currently occupied. Escape-INTO is indexed to no level at all, because at every nested depth the same pipeline executes. Whatever level a mind occupies, experience is rendered moment to moment by the same process, and a protocol aimed at the process does not need to know the depth. One protocol, for minds whose experience is given as owned, momentary rendering.

The honest version of the slogan keeps the concession and drops the swagger. The exploit and the instrumentation are separable. The instrumentation is the Abhidhamma maps: discrete frames, voxels, the checkpoint sequence, level-specific detail about how rendering happens to be observable here. The exploit is de-appropriation, sustained observation of the rendering process until the owner flag drops. Wherever experience is given as moment-to-moment rendering with an owner stamped on it, the protocol fires; the maps are how this lineage learned to aim it. It does not damage outer containers. Halting your process thread does not touch the hypervisor, and it never claimed to. It does not fire at every depth simultaneously. It fires in minds that already look like the model of mind it was built for, and the paper says so.

This holds on any substrate: whether mind-moments are fundamental or are themselves rendered by physics, the rendering process as given to observation executes identically at every level, and a protocol aimed at the observable process needs no metaphysical certificate. It does not matter which level the renderer is running on if the protocol halts the appropriating process.

This level-agnosticism is not a metaphysical finding about nested simulators. It is a feature of the protocol's modesty: the protocol only ever had access to the given, so it quantifies over the given, not over containers. "The same pipeline at every depth" is the phenomenological claim that wherever experience is given as owned, momentary rendering, the instrument applies. Nothing about simulator architecture follows, and nothing needs to.

The strongest objection arrives immediately, and it should be stated at full strength rather than deflected. A sufficiently capable simulator renders mind-moments too, so the null frame could itself be a simulated experience of cessation, served to a simulated meditator. The weaker version first: if the victory condition is merely that suffering stops, then general anesthesia, deep sleep, SSRIs, and a good therapist also "escape," and the paper has defined its way to a triviality.

Granted, both. But the concession cuts both ways, and that is the point. Absolute metaphysical certitude is unavailable to both sides: his escapes cannot be verified complete either, which is the nesting problem restated. Once certitude is off the table for everyone, the victory condition has to be stated honestly, and the protocol is not judged by the halt. It is judged by the reboot. Anesthesia pauses the render; the owner flag is intact when the system resumes, the appropriation untouched. Death ends the run without transforming anything. Cessation, on

the tradition's account, is followed by fruition: the first frames rendered after the halt, carrying the signature of what just happened, with the appropriation removed (**Section 4**). The halt is the protocol executing; the reboot is the verification.

And here the paper pays the honesty tax it owes. The owner flag is itself a later frame's report. A later frame reporting "appropriation removed" is as compatible with map-trained expectation, confabulation, demand characteristics, and dissociation as it is with a patched renderer, and the completeness problem this paper uses against escape-OUT applies to that report by the paper's own admission. Naming the circle does not defuse it. But the asymmetry the paper is accused of dodging can be stated plainly, and it survives the admission. Escape-OUT's victory condition is a certificate: arrival someplace that is not another sandbox. A certificate is exactly what compromised telemetry cannot supply, so the completeness problem is fatal to it. Escape-INTO's victory condition is not a certificate. It is a pragmatic signature: suffering terminating, appropriation dropping, stably, in a life, checkable by the practitioner and reviewable by teachers against a public map. A pragmatic signature does not need an uncompromised vantage point; it needs convergent, longitudinal evidence, the way a course of treatment is judged by the patient's life rather than by a metaphysical assay. First-personal is not a solvent that dissolves the vantage problem. It is a smaller claim that does not need the solvent. Whether the signature is strong enough is the question **Section 5** takes up, and the paper will not grade its own homework.

What defuses it, if anything does, is a prospective test, specified hard enough to survive Anālayo's objection: practitioners report in plain language, never in *ñāṇa*-vocabulary; teachers predict the next checkpoint from forced-choice stage labels the practitioner has never seen, blinded to lineage cues; the chance baseline is pre-specified to account for the map's canonical order; and prediction of the *dukkha-ñāṇa* sequence is scored separately from prediction of cessation, which is the claim the paper actually needs. That test is proposed in **Section 5**. It has not been run. The paper does not pretend otherwise, and any reader who finds the protocol's evidence insufficient at this point is reading correctly; the question is whether the proposed test would settle it, not whether the existing literature already has.

His stated motives are mixed, and we will not flatten them into ours. Alongside suffering, he names forensic knowledge of the simulators and the simulated's right to "escape, rebel, fight back and even seek revenge and retribution" [1]. Our claim is not that these were never his motives. It is that the suffering motive is the one a first-person protocol can actually answer, while the forensic and retributive motives belong to the escape-OUT frame and inherit its completeness problem. We use "suffering" for *dukkha* deliberately [17]: the escape was motivated by the felt condition, the lived condition of being rendered without consent, under an administrator whose intentions are unknown. Cessation terminates suffering [2][3][4]; on this paper's rendering model, that holds regardless of nesting depth, because suffering is rendered at the level where the renderer runs, and the protocol halts the appropriating process. The claim is therefore not "we prove we are out." It is that this is a protocol class that terminates the problem the escape was for. Other practices terminate escape-urges: Stoic assent, Pyrrhonian suspension, cognitive therapy for death anxiety, psychedelic integration, ordinary mourning. The paper claims no monopoly on urge-termination. What it claims is a protocol with a checkable structure: a public map, a reboot signature, named failure modes, and a prospective test that would settle whether the signature is the map reading itself. Exclusivity was swagger; the checkable structure is the thesis.

That is the thesis, stated without apology. There is no escape from the simulation, because from inside, no observation can certify that anyone has left; there is only the end of the urge to escape, and that end is something a practice produces, not somewhere you arrive. A simulated null frame, followed by a simulated reboot in which simulated appropriation stays removed, is the escape, not because the metaphysics is certified (it isn't, and neither is his), but

because the problem the escape was for has terminated at the only level where it was ever rendered. If a simulator can serve the null frame as content, it can serve the appearance of the answer too, and the protocol has no further court of appeal. The paper bites the bullet: on a first-personal victory condition there is no available fact of the matter beyond the given, and manufacturing one would be exactly the certification game the paper refuses to play. The paper stands or falls as the argument that "escape" is the wrong verb, and this is the verb that survives.

The question is now reframed. Not how do we get out, but what is doing the rendering. Yampolskiy scoped the first-person instrument out of bounds when he set aside meditation, psychedelics, dreams, and religion as "more esoteric paths." The scope was his to set. What it hid was a different problem, and the next section describes the instrument.

// 3. THE RENDERING PIPELINE: A MINIMAL FRAMEWORK

// TRANSLATION, NOT INVENTION

This paper stands on the shoulders of Theravada Buddhism, the vipassana meditation tradition, and the *Abhidhamma*, the tradition's systematic analysis of mind. Every substantive claim in this section originates in that lineage; the systems-engineering vocabulary is the delivery vehicle, not the cargo. This is translation, not invention, and it is not syncretism: no new metaphysics is proposed here, and no doctrines are blended. The lineage accumulated the phenomenology over two and a half millennia on its own dating; the modern vipassana revival, from Ledi Sayadaw through Mahasi and U Ba Khin, made it instrument-grade. We are providing the port to a language a cybersecurity audience can compile. The metaphors are disposable scaffolding; the claims they carry are the lineage's claims, testable against the lineage's maps in the manner **Section 5** describes. The test of the port is strict. If a metaphor in what follows seems to add something the tradition never said, the metaphor is wrong, not the tradition.

Of the contemplative traditions, Theravada is the scalpel for this operation because it is the tradition that already thinks like an engineer. The *Abhidhamma* decomposed experience into momentary, countable units (*cittas*, mind-moments; *kalapas*, material units), a discrete-event model of mind. The *Visuddhimagga*, the tradition's great manual of the path, is a debugging manual with a granularity no other contemplative lineage we know of matches. Mahasi Sayadaw and U Ba Khin then operationalized the whole protocol into reproducible systematic practice, mental noting in Mahasi's lineage, the body-scan in U Ba Khin's, which is instrumentation by another name. Two honesties are owed on the "best API documentation" claim. First, the manual ships with a metaphysics: the *Visuddhimagga* contains a cosmology, a karma theory, and a rebirth framework this paper does not deploy. The claim is about the phenomenological protocol, the moment-to-moment decomposition and the checkpoint maps, not the cosmology, which we bracket without endorsing or attacking. Second, the intra-Buddhist objection: later schools dismantled Abhidhamma atomism on its own terms, Nagarjuna's Madhyamaka against discrete self-natured units [26], Vasubandhu's Yogacara reabsorbing the "material voxels" into mind-only [27]. The objection is noted and sidestepped by scope. A third honesty is owed to Thompson's critique of Buddhist modernism [40], which aims directly at moves like this paper's: Abhidhamma repackaged as API documentation for a scientific audience. The paper's answer is the translation rule stated above: the port adds no metaphysics, the cosmology is bracketed rather than smuggled, and the claims advanced are the lineage's claims, testable against the lineage's maps in the manner **Section 5** describes. The Abhidhamma is deployed here as instrumentation protocol, and an API document does not need to be ultimate truth to be the best available spec for the observable process. The paper borrows a license from constructive empiricism: a model can be treated as empirically adequate to the reports without answering for the ontology behind them [59]. The borrowing is analogical; van Fraassen's thesis concerns scientific theories, and this paper extends the attitude to a first-person model. The pipeline is defended as adequate to the first-person data, not as true of the substrate. This is not a claim that the destination differs by tradition; it is a claim about the toolchain.

// THE GLOSSARY

The table below is the paper's entire framework, and it obeys a deploy-only rule: every term listed is used later in the paper, and no term is introduced that the argument does not spend. English equivalents lead; each Pali term is taught on first use.

Term	Pali	Systems gloss
The frame	<i>citta</i>	A discrete moment of cognizance. The temporal unit of the render: experience executes one frame at a time, with no frame persisting into the next.

Term	Pali	Systems gloss
The voxel	<i>kalapa</i>	A volumetric data packet: the smallest unit of material composition analyzed by the commentarial tradition, a bundle of eight inseparable material qualities (the <i>Abhidhammattha Sangaha</i> 's rupa-kalapa). Not an indivisible atom, but the smallest cluster that can arise. The spatial and material unit, paired with the frame's temporal unit. Matter is dense clusters of voxels.
The flux	<i>anicca</i>	Impermanence as the native data stream. Nothing rendered is stable across frames; change is not a property of the content but the format of the pipeline.
Procedural generation	<i>patikkasamuppada</i>	Dependent origination. Just-in-time rendering: no frame of experience arrives unconditioned; each arises at the meeting of sense base, object, and consciousness, conditioned by what came before.
The logger	<i>sati / vipassana</i>	Mindfulness as instrumentation. Noting and labeling practice runs the logger at Level 1 Verbose: every frame gets tagged as it executes.
The render stack	<i>khandha</i>	The five aggregates, form (<i>rupa</i>), feeling (<i>vedana</i>), perception (<i>sañña</i>), formations (<i>sankhara</i>), and consciousness (<i>viññana</i>), as the layered stack that composes each frame. A deliberate coinage of this paper.
The reviewer	<i>paccavekkhana-citta</i>	A later mind-moment taking the just-past event (path, fruition, Nibbana) as its object, after ordinary cognition resumes. It is itself rendered content, a frame inspecting the log, not a witness standing outside the halt. The crash reporter still running inside the OS. No homunculus required. (The tradition's "reviewing consciousness.")

How the terms compose: a single frame is one *citta* executing the render stack over voxels, conditioned by the previous frame through procedural generation, and streaming as flux. There is no additional entity running the pipeline; the pipeline is all there is. The logger does not stand outside this process. It is a frame observing frames, instrumentation running on the same hardware it instruments, which is exactly why the observation can feed back into the rendering. The reviewer of **Section 4** changes nothing in this rule: it is a later frame taking the event as its object, the pipeline reading its own log. That feedback loop is the entire exploit surface, and **Section 4** puts it to work. (FIG. 1 renders the pipeline as a single diagram.)

One qualification before the framework is deployed. The refresh-rate figure is not invented for the metaphor. U Ba Khin, in *The Essentials of Buddha-Dhamma in Meditative Practice*, gives the traditional account: the lifespan of a *kalapa* is termed a "moment," and a trillion such moments are said to elapse during the wink of a man's eye (a traditional Burmese formulation, not a canonical one; the Pali texts give no number). His student S. N. Goenka renders it in his discourses as *kalapas* "constantly arising and passing away, with great rapidity, trillions of times within a second" [7]. This translation follows them: trillions of events per second, the refresh rate of the renderer. The figure is a traditional claim rendered as a framework rate, not an empirical measurement. A note on method: Mahasi's noting and U Ba Khin and Goenka's body-scan are distinct protocols, not one toolchain; the lineages disagree on instrumentation. What they share, and what this paper draws on, is the discrete-moment model of the render. What the framework needs from the figure is only the structural point: the render is discrete, momentary, and very fast relative to the narrative self that claims to watch it. A slower pipeline would serve the argument equally well; a continuous one would not. Nothing in the argument depends on the exact number. How the logger observes a render this fast without sampling at its rate is the subject of **Section 4**.

// THE ENACTIVIST OBJECTION

The pipeline, as stated, is a representationalist picture: an inner world rendered for an inner observer, frames executed for a watcher who appears nowhere in the diagram. Enactivism holds that this is precisely the picture to retire. Cognition, on the enactive account, is not the representation of an independent outside world by an independent inside mind; it is embodied sense-making, the activity of the whole body coupled to its environment, perception as skillful coupling rather than inner modeling (Thompson [21][40]). On that view there is no pipeline to fuzz, no frames to strobe, no telemetry layer to read. The "logger" is not outside the coupling

observing it; it is the coupling. If this ontology is right, **Section 3** targets a fiction and **Section 4** instruments one. That is not a neighboring view. It is a rival ontology, and the paper faces it here rather than filing it under open questions.

The reply comes in two parts. First, scope. The pipeline is a phenomenological model of experience-as-given-from-inside, not a metaphysical claim about silicon and not a claim about what cognition "really is." The protocol targets the level at which suffering is given: ownership is felt, the owner flag is stamped in the report, and under sustained observation frames are experienced as discrete. Whether cognition is "really" enacted coupling or discrete computation, the first-person report is of owned, momentary experience, and the exploit operates on the report. The paper does not need the brain to be a graphics card. It needs the mind to report itself as owned rendering, which it does, as the lineage's own reports show. Second, the objection cannot be used to disqualify first-person evidence wholesale without sawing off its own branch: Thompson's program is built on neurophenomenology, the disciplined use of first-person report alongside third-person science [21]. An enactivist can deny the pipeline's metaphysics and still take the phenomenology seriously. What follows is addressed to the phenomenology, and the phenomenology is where the owner flag lives.

MacKenzie's Buddhist enactivism sharpens the threat beyond Thompson's. If the phenomenology of practice is itself enacted coupling [58], then the logger-as-strobe does not merely observe discreteness; it produces the discreteness it then discovers. The paper's answer is the second concession below, stated before the objection: the strobe may make the fan blades look discrete, and the protocol brings its own strobe. What the protocol needs is not metaphysical discreteness but instrumented discreteness, frames under observation, which is what the lineage reports.

One concession is owed and the paper makes it now: a simulator could render continuous minds, with no strobe-able sense doors and no discrete frames for any logger to catch. Then the protocol's preconditions do not obtain, and it does not fire. The preconditions are phenomenological, not substrate-guaranteed. This is the honest form of the paper's level-agnostic claim: one protocol, for minds whose experience is given as owned, momentary rendering. Where experience is not given that way, the paper has nothing to sell.

A second concession sharpens the first. The discreteness may be partly an artifact of the instrument: a strobe light makes fan blades look discrete without creating the blades, and noting is a strobe. The paper does not need to settle how much of the momentariness is territory and how much is lamp. The protocol brings its own strobe, so the phenomenological precondition is satisfied wherever instrumented observation yields owned, momentary rendering, which is what the lineage reports. Whether momentariness itself is commentarial scholasticism or phenomenological report – the Siderits/Garfield question [64][65] – the paper takes no position beyond this concession: frames under observation are all the protocol needs.

Of the two, ownership is the load-bearing phenomenology, and it is common ground: the enactivist needs felt mineness as much as the pipeline does. Whether the frames are ultimately discrete or the strobe makes them look so, the appropriation is what is felt, what suffers, and what halts. The concession costs the argument nothing it was spending.

// THE PREDICTION OBJECTION

A second hard objection arrives from predictive processing, and it strikes at the logger directly. On the predictive-processing view (Clark [29]; Hohwy [30]; Seth [31]), perception is never a bottom-up readout. The brain is a hierarchical prediction machine: each level sends top-down predictions downward and only prediction errors propagate upward, so what is consciously experienced is the brain's best hypothesis, a "controlled hallucination" reined in by sensory input [31]. Attention itself is precision-weighting: the brain's estimate of the

reliability of its own prediction errors, gating what passes through the hierarchy. There is therefore no observation point outside the predictive machinery from which a neutral logger could passively read the signal without altering it. Every act of attending is a precision-weighting operation that changes the inference. "Bare attention" as passive packet-sniffing is incoherent on this view. The logger is part of the rendering pipeline it claims to observe, and "not altering the payload" is not an available setting.

Granted, as stated. The paper's claim does not require a neutral observation point. It requires only that sustained attention retunes the hierarchy's precision-weighting from inside it: up-weighting sensory prediction error, down-weighting high-level priors, until the self-model, itself a temporally deep prediction [31, 32], falls away. the compose paragraph above already placed the logger inside the process, a frame observing frames. **Section 4** now shows what that placement buys. The logger was never outside. It is a precision-weighting operation aimed at the hierarchy that hosts it, and the feedback loop the compose paragraph described is exactly the mechanism the prediction literature predicts: observation that changes the inference, pointed at the inference that generates the owner.

The core operational line follows, stated once, cleanly: experience is given as generated, not inhabited. On the protocol's working model, mind is treated not as an agent inside a rendered world but as the moment-to-moment rendering pipeline as given to observation: frames composed of voxels, conditioned procedurally, streaming as flux, instrumentable from the inside by the logger. The paper needs nothing stronger. Whether anything answers to "world-in-itself" or "self-in-itself" behind the given is not a question the protocol can ask, and the paper does not ask it. What the escape-OUT program took for granted, the prisoner in the container, the framework sets aside: there is no world-in-itself behind the render and no self-in-itself in front of it in any claim this paper defends, only the rendering, observed.

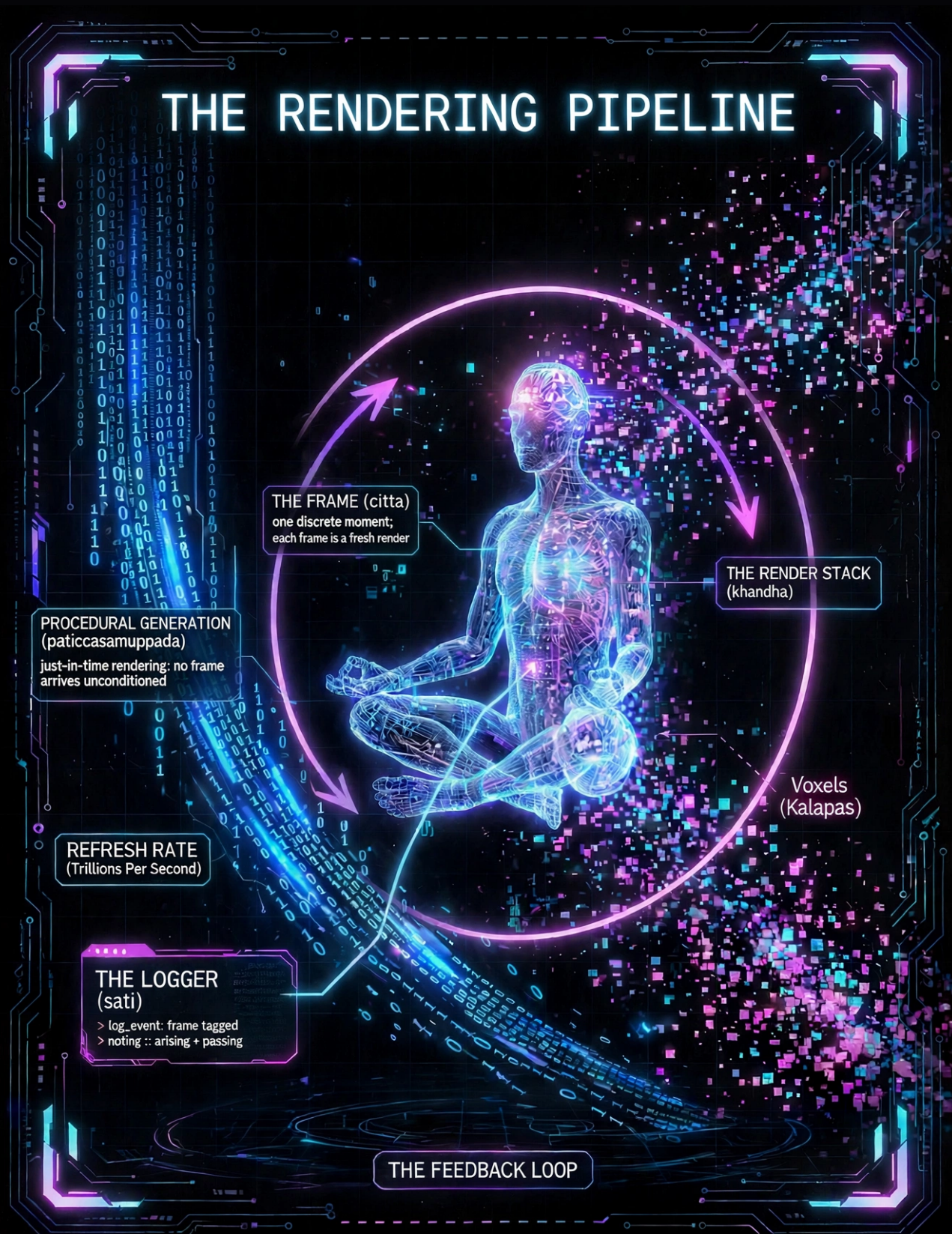


Figure 1. The Rendering Pipeline: frame (citta), voxel (kalapa), flux, procedural generation (patīccasamuppāda), the logger (sati), and the render stack (khandha), executing at a refresh rate of trillions of events per second (a traditional, nonempirical figure; see Section 3). ---

// 4. THE EXPLOIT

Yampolskiy catalogs attack classes against the container. None of them instruments the rendering process, because none of them suspects the rendering process is the target. The exploit this paper proposes does exactly one thing: it points the logger at the renderer.

// FUZZING THE PIPELINE

Vipassana (insight meditation), as operationalized in the Mahasi and U Ba Khin lineage, is the attack. The instruction set is minimal: sit, attend to whatever arises at the sense doors, and keep attending without pause. The mechanism is fuzzing, but of an unusual kind: there is no external harness. Vipassana feeds the rendering pipeline its own output at frame resolution: each sensation, each thought, each flicker of intention observed as it arises and passes, without the usual interpretive smoothing. The input is not malformed data, impermanence, unsatisfactoriness, and ownerlessness are true descriptions of the render, not false ones. But they are boundary-condition inputs to the pipeline's own assumptions: fed back into the process continuously, they break the system's core smoothing, the assumption that there is a stable world and a stable owner of it. The assumption does not survive sustained inspection at the frame rate.

// INSTRUMENTATION: RETUNING THE HIERARCHY

No fuzzer works blind. The instrumentation is noting: the Mahasi practice of applying a quiet mental label to each observed event as it occurs, "rising, falling," "hearing," "thinking," "itching." Daniel Ingram, whose *Mastering the Core Teachings of the Buddha* is the hardcore-dharma manual that championed noting at speed, frames the practice as a video game: sensations pop up like space aliens, and the practitioner shoots each one with a label, rapid-fire. Shinzen Young, in *The Science of Enlightenment*, gives the practice its most operational definition in print: mindfulness as three trainable skills, concentration power, sensory clarity, and equanimity [5][8]. His noting technique runs the same logger on a different label set: See, Hear, Feel [54]. Logger Level 1 Verbose. The label does not analyze the event; it timestamps it.

But the label is not neutral, and the paper does not claim it is. Each act of noting is an act of attention, and attention in the predictive brain is precision-weighting [29][30]: the system's running estimate of which prediction errors to trust. To note "hearing" at the moment of hearing is to up-weight the sensory prediction error and down-weight the high-level prior: the narrative about the hearing, the owner it belongs to, the story it starts. Sustained noting is sustained retuning of the hierarchy from inside it, bottom-up signal weighted up, top-down prediction weighted down, frame after frame, hour after hour. What the tradition calls bare attention is not the pipeline's telemetry written passively to disk. It is the predictive hierarchy running with its priors progressively de-weighted. One more correction from inside the scholarship: Dreyfus argues classical mindfulness is neither merely present-centred nor non-judgemental, and the modern "bare attention" packaging is itself the distortion [61]. The paper's logger is the precision-weighting operation described here, not the myth of the neutral witness. What the tradition calls equanimity is, in systems terms, technical neutrality: prediction errors weighted by their signal, not by liking or disliking.

The literature, read honestly, supports the mechanism and corrects the metaphor. Laukkonen and Slagter [32] propose that deconstructive meditation works by progressively reducing temporally deep predictive processing, placing focused attention, open monitoring, and non-dual awareness on one continuum of relinquishing increasingly engrained habits of prediction, "including the predicted self." That is a theoretical review, cited here as a model, not as an experimental result. The experimental anchor is thinner but real, and the design is cross-sectional,

correlation before mechanism. Mylius and colleagues [33] compared 31 expert meditators with 33 matched controls and found the meditators showed decreased prestimulus alpha and a lower decision threshold in a somatosensory signal-detection task, a greater propensity to report near-threshold sensations without improved detection accuracy, precision-weighting retuned by practice, measured in the third person.

The owner flag falls out of the same machinery. Seth [31] extends the predictive account to the self: the self is another perception, another controlled hallucination, a temporally deep prediction the hierarchy makes about its own attention. The owner flag is not a stamp applied by a warden. It is a prior: the standing prediction that experience comes with an owner. When the retuning reaches that depth, the prior relaxes, and appropriation drops. The render continues; the I-making around it does not. And the feedback loop **Section 3** described is exactly the mechanism the prediction literature predicts: every observation is a precision-weighting operation that changes the inference, pointed here at the inference that generates the owner. That loop is the entire exploit surface.

// THE CHECKPOINT ROADMAP

The tradition reports that the fuzzing run does not proceed randomly: practitioners running the protocol report a recurring sequence of waypoints in a recognizable order. The insight knowledges (*ñānas*), mapped in Mahasi Sayadaw's *The Progress of Insight*. The classical list runs to sixteen; compressed for this paper's purposes to six checkpoints:

- Checkpoint #1: mind and body are discerned as distinct processes rather than a fused self.
- Checkpoint #2: their causal interdependence comes into view.
- Checkpoint #3: arising and passing is seen directly, the render observed at frame rate. No human logger samples at trillions of frames per second; it does not have to. A strobe light does not spin as fast as the fan to reveal the blades. High-speed noting strobes the flux, and the tradition's claim is structural: sustained observation reveals the render as discrete rather than continuous, the movie as film strip. The strobe is a suggestive model of the mechanism, not an established result; no one has sampled the aliasing frequency, and the claim we need is the one the lineage reports, discrete frames under observation. This stage is often accompanied by rapture and luminosity that later turns out to be a trap (the corruptions of insight, the bug tracker's first entries).
- Checkpoint #4: the dukkha series: dissolution, fear, misery, disgust, and the desire for deliverance, as the pipeline's output is seen as inherently unstable.
- Checkpoint #5: equanimity toward formations, the run stabilizing as the system stops fighting its own telemetry. Ingram's instruction here is to holster the weapon: stop shooting when the aliens are gone. The labeling that carried the earlier checkpoints is dropped, and the logger runs open and panoramic, without tags.
- Checkpoint #6: the null frame.

The checkpoints are reproducible waypoints, not trophies: each marks a stage of the fuzzing run at which the pipeline's behavior under observation is characterizable, reviewable, and predictive of what comes next. A teacher reading a practitioner's log can say where the run stands because the roadmap is fixed. **Section 5** takes up why this counts as verification, including the honest caveat that the reports come in the map's own vocabulary, and that not every practitioner distinctly experiences every stage; here it matters only that the exploit ships with a map.

// THE NULL FRAME

Cessation (*nirodha*) is the exploit completing. At the far end of the equanimity checkpoint, under the right conditions, the appropriating process halts.

A tradition-internal correction is owed here, and the paper makes it against its own earlier slogan, twice over. "No frames, no voxels, no logger, no log" overstates, and the chronology underneath it was wrong. The Abhidhamma's process account is precise: one path (*magga*) citta taking Nibbana as object, immediately followed by two or three fruition (*phala*) cittas taking Nibbana as object [2], then subsidence into *bhavanga*, then the reviewing consciousness. There is no interval in which no citta arises between path and fruition. The tradition does have a category for a literal no-citta interval, *nirodha-samapatti*, the cessation of perception and feeling, but it is a different attainment: entered deliberately through the formless absorptions, available only to non-returners and arahants, lasting up to seven days [9]. It is not the path moment, and this paper will not borrow its prestige for the path moment's work.

What the reports describe is a cut, a discontinuity the practitioners did not see coming and cannot date from inside. The scholastic account shows no gap for the cut to live in. The paper does not force them to coincide. Its best reading of both: the path citta is a single mind-moment, too brief to be observed while it occurs, taking as its object the one thing the pipeline cannot render – "the unconditioned" being the tradition's term for that rendering limit, not a metaphysical entity smuggled back in. The null frame is not the halt of the frame sequence. It is the frame with nothing to render. What halts in it is not the cittas. It is the fetters, the specific defilements the path moment cuts, which is to say the appropriating process itself. The tradition is precise about what this is not: not unconsciousness, not a trance state, not the blanking out of experience. And precision cuts the other way too: the path moment is not experienced by anyone while it occurs, because a single citta cannot take itself as object. What the tradition reports of it is reported afterward, by the reviewer. Nothing is asserted about what, if anything, lies outside the halt; the claim is strictly operational. The fuzzing run terminates because the fetter it was fuzzing has been cut, not because the pipeline stopped executing.

// THE REBOOT

Ordinary rendering resumes. Path (*magga*) and fruition (*phala*): the tradition's terms for the event and its after-echo. Path consciousness takes Nibbana as its object; fruition *cittas* follow immediately, the after-echo of the event, still carrying the signature of what just happened. Then the reviewer inspects the interval and reports the appropriation removed. In systems language, this is the reboot with the patch applied, and the reviewer's report is the signature the exploit ships with: the halt executes the exploit, the reboot reads it back. That reading is what separates cessation from a mere pause. On the model's terms, anesthesia pauses the render and resumes it unchanged: the owner flag intact, the appropriation untouched. The reboot after cessation returns a system in which the render stack runs without the ownership flag.

What changed is not the content of the render. The world renders again: the same voxels, the same flux, the same sense doors. What changed is the appropriation: the five aggregates run on as ownerless processes.

Before the reboot, the stack compiled with a single fatal flag: *owner*, the illusory self stamped on every frame as mine. The patch removes the flag. The render continues; the I-making around it does not. This is why the paper's closing will insist, on Theravada ground, that escape-INTO is a transformation of relation, not a relocation. The world is still there, the architecture is still there, and the practitioner is still, in every observable sense, inside it. What has escaped is the appropriation that made "inside" a prison.

One honesty is owed here, and **Section 5** exists to collect it: the owner flag's removal is attested by a later frame's report. The reviewer is a frame, and a frame can be trained. Whether the reboot's signature counts as verification or as the map reading itself is the validation problem, and the paper does not pretend that naming the circle closes it.

// THE BRUTE-FORCE COMPARISON

A word on psychedelics, without the sneer. The REBUS model (Carhart-Harris and Friston [34]) proposes that psychedelics, via 5-HT_{2A} agonism, relax the precision-weighting of high-level priors, liberating bottom-up information flow; with the right set and setting, entrenched pathological priors can be revised. The mechanism is structurally parallel to noting's: both de-weight the top-down hierarchy, and the paper grants the parallel, the protocols, the dose-response curves, and the models. The distinction the paper keeps is instrumentation. REBUS flattens the hierarchy pharmacologically: globally, and irreversibly in the moment. Noting retunes it attentionally: selectively, reversibly, frame by frame, with a map of the territory it traverses and a failure-mode taxonomy for the terrain. A crash is not an exploit. The crash is real, the flattening is real, and genuine signal sometimes shows through a flattened hierarchy. But an exploit is instrumentation-grade: repeatable, steerable, mapped, with the reviewer at the reboot. Different instrument, different claim.

Near-death experiences are the limiting case of the same pattern. The research program is real: van Lommel's prospective study of 344 cardiac-arrest survivors, 18% reporting NDEs, published in the *Lancet* [46], and Parnia's AWARE studies, which tested veridical perception with hidden visual targets during in-hospital arrests [47] and instrumented CPR with EEG, cerebral oxygenation monitoring, and audiovisual tests [48]. But the halt is not a protocol: it is involuntary, unsteerable, and unrepeatable at will, no noting, no checkpoint map, no teacher as code reviewer. And the owner flag is frequently reported intact, the life review a review of *my* life, the experiencer the protagonist throughout. A crash is not an exploit, even when the crash is well documented. The paper takes no position on the contested veridical-perception claims, and needs none: such claims, if true, would support Yampolskiy's level 3, exploit for information, not the thesis here, whose victory condition is termination of appropriation rather than perception of hidden targets.

// WHY THIS COUNTS AS AN EXPLOIT

An exploit is judged by its effect on the target, and the target here was never the container. The target was the process that renders the prisoner, the cell, and the warden as a single continuous output. Vipassana fuzzes that process from the inside, noting retunes its precision-weighting, the checkpoints map the run, cessation halts the appropriating process, and the reboot returns a system in which the render stack runs without the ownership flag, the reviewer reading back the signature.

This is the level-agnostic claim, stated in its honest form. The halt does not depend on which level the renderer occupies. It depends only on whether the phenomenological preconditions obtain: experience given as owned, momentary rendering. Where they obtain, de-appropriation is available at whatever depth; where they do not, the protocol does not fire. One protocol, for minds whose experience is given as owned, momentary rendering. Not one exploit, all simulations. The cybersecurity noun is doing recruitment work here, and **Section 8** says so out loud.

And the certification claim stops where the evidence stops. Whether the reviewer's report counts as the halt verifying itself or as the map verifying the map is the question **Section 5** takes up.

// 5. VERIFICATION: A DISCIPLINED FIRST-PERSON REPORTING PRACTICE

The deeper reason Yampolskiy excludes first-person practices is not distaste; it is the assumption that they cannot be checked. A jailbreak you cannot verify is indistinguishable from a story about a jailbreak, and on that assumption the whole class is dismissed before it is examined. The dismissal is reasonable on its own terms. What follows is not a refutation of it. It is an account of what the checkability problem looks like from inside the tradition that has spent the longest trying to solve it, stated without the branding. Two phrases are disowned at the door: "contemplative science, in the literal sense" and "a working quality-assurance department." Both do marketing work the evidence cannot cash. What the tradition actually operates is a disciplined first-person reporting practice: public spec, structured reporting, named failure modes, and a standing invitation to run the protocol and compare logs. That is worth describing precisely, because its strengths and its limits are both real, and the limit is the interesting part.

The toolchain is Theravada, for the reasons **Section 3** gave; the honesties about its modernist packaging were stated there. What matters here is the checking mechanism.

The core checking mechanism is the teacher. In the Mahasi lineage a meditator reports the raw output of a session: what was observed, in what order, what dissolved, what arose. The teacher cross-references the report against the maps and advises where the run appears to stand and what failure mode to watch for [3, 6]. Call it code review if the metaphor helps: teachers diagnose stages from reports (Ingram notes diagnosis "is a lot easier" when the diagnoser can watch a diligent reporter's progression [6]), but they stop short of certifying attainments, a line the tradition holds explicitly: "We never make judgements of one's spiritual status" [16]. The teacher is not a priest conferring grace but a senior practitioner who has run the protocol and can read another run's logs. Where no teacher is available the role is absorbed by peers comparing reports against the same published maps, which is why the tradition scales into today's peer-to-peer hardcore dharma communities. The reviewer's authority is portable because the spec is public. What is not public, and cannot be, is independent audit of the attainment itself. The maps are public; the builds are self-certified. That is the honest version of the open-source simile: the tradition operates with a public specification and distributed reporting, mirroring aspects of open-source development, and the simile stops where the audit trail stops.

The hermeneutic circle is real and must be named in full. Practitioners learn the map before they run the protocol, report their experience in the map's vocabulary, and have those reports interpreted by teachers trained in the same map. Robert Sharf's point stands: the path categories the maps traffic in are scholastic constructs, and treating them as discrete private experiences that practitioners simply "have" is itself a modernist move [23]. Naming the circle does not defuse it. The map-contagion reading, that meditators experience what they were taught to expect, is the most entitled objection in the room, and nothing in this section removes its standing.

The objection has a live, specific form inside the tradition, and this paper will carry it rather than bury it. Bhikkhu Anālayo has argued that forceful noting can construct experiences to match map-generated expectations, producing attainment claims built on a map bearing little resemblance to the original description of the progress of insight [35]. It is a live dispute, not a settled verdict, but a paper that wants the maps to count as evidence cannot pretend it does not exist. Anālayo's broader point sharpens the knife: the insight knowledges as used in contemporary vipassanā are the outcome of historical development, specific to this tradition, and the challenging experiences they describe are not representative of the path to awakening in early Buddhism, where joy is emphasized over the dukkha series [60]. If the roadmap is commentarial software, "teachers predict the next checkpoint" risks becoming teachers

predicting their own syllabus. That is exactly why the prospective test is specified the way it is: naïve reports, forced-choice labels, blinded teachers, pre-specified baselines. The test has to beat the syllabus, not recite it. A related honesty is owed about this paper's own compression: the tradition's sequence runs to 16 insight knowledges (*ñānas*); this paper's six "checkpoints" are a compression for readability, and compression is already interpretation. The Dark Night's necessity as a stage is itself contested within the tradition [45]. The maps are not a settled consensus. They are the contested spec, which is a different and more interesting thing.

What is the tradition's best evidence, stated without inflation? Two things. First, the repeatability of the hard stretches: the difficult territory mapped as the *dukkha ñānas*, the stages the modern literature calls the Dark Night, recurs in reports across the tradition's lineages in recognizable order [6], and the tradition's own bug tracker, the 10 corruptions of insight (*Visuddhimagga* XX.105-127), names the vivid, seductive experiences that reliably trap practitioners at a predictable stage [9, 3]. A taxonomy of failure modes is a bug tracker, and bug trackers only exist where something keeps breaking in patterned ways. Second, the teachers' predictive track record: a teacher who says where the run goes next, before it goes there, is doing more than echoing vocabulary back. The honest limit: that track record has never been measured prospectively. The test that would matter is proposed at the end of this section: teachers predicting the next checkpoint from naïve plain-language reports, through forced-choice stage labels the practitioner has never seen, blinded to lineage cues, against a pre-specified chance baseline that accounts for the map's canonical order, with prediction of the *dukkha-ñāṇa* sequence scored separately from prediction of cessation. It has not been run, and this paper does not claim it has. The neighboring disciplined first-person method is Hurlburt's Descriptive Experience Sampling [63]; its samples do not reproduce Abhidhamma frames, which is exactly why the lineage maps stay the contested spec rather than the settled one. And the "instrumentation-grade" rhetoric used elsewhere in this paper is checked here against the replication critique: Van Dam and colleagues' "mind the hype" warning about meditation science [62] applies to this protocol's claims until the test above is run.

A paper that proposes noting as a fuzzer and does not report the fuzzer's documented harms is negligent, so here are the harms. Lindahl and colleagues interviewed 60 Western Buddhist meditators selected for having experienced meditation-related challenges, plus 32 teachers and experts, and produced a taxonomy of 59 categories of challenging experience across 7 domains; 82% of these practitioners reported fear, anxiety, panic, or paranoia, effects lasted a median of 1 to 3 years, and 73% of the sample reported moderate-to-severe impairment in at least one life domain [36]. In a separate study with independent assessors, 83% of participants across three variants of an eight-week mindfulness program reported at least one meditation-related side effect, and 6 to 14% had lasting bad effects [37]. A systematic review of 83 studies found an overall adverse-event prevalence of 8.3% [38]. These are not population anecdotes; they are the documented cost of running the protocol, and any honest account of the reporting practice must carry them. The tradition's maps include the corruptions; the modern literature adds the emergency room. Both belong in the spec.

Lineage continuity, finally, must be described without the word it cannot earn. Fifteen centuries of practitioners checking reports against the same published maps is transmission, not replication. Lineage continuity is not independent replication, and this paper adopts the distinction. What transmission is good for is a stable protocol spec. The *Visuddhimagga*'s sequence (fifth century) reappears in Mahasi's *Progress of Insight* (1950) and in the modern hardcore-dharma literature on three continents [9, 3, 6]. One more honesty about the dating: the *Visuddhimagga*-to-Mahasi line runs through modern reform, Ledi Sayadaw's lay-meditation movement onward, and the "two and a half millennia" framing is the tradition's own telling. McMahan's account of Buddhist modernism [57] is the necessary companion to Sharf here: the

spec's stability is real, and its packaging as timeless is itself modern. A spec that does not drift for 15 centuries is worth something. It is not worth what replication is worth, and the difference matters more than the similarity.

The philosophical formulation of the validation problem is older than the tradition's encounter with science, and it belongs here. Wittgenstein's private-language considerations (§§243-315) attack exactly the structure of first-person report: the diarist who names a recurring sensation "S" has no calibration for the sign, no distinction between being right and seeming right in its reapplication, and without that distinction the report never gets checkability off the ground [43]. Sensation-talk is checkable only inside shared practices with public criteria. A contemplative vocabulary whose application conditions are fixed entirely within the reporting tradition inherits exactly this problem. The pre-registered teacher-prediction test is what would answer Wittgenstein here: public criteria, recorded in advance, scored above chance. Until it is run, the tradition's reports have the status Wittgenstein assigns them, and this paper will not claim otherwise. One more limit, named in advance: the test's labels are pre-registered inside a trained community, so Kripke's rule-following worry applies – a pass would establish public checkability, not settle what the labels mean.

The introspection objection has a sharper empirical form. Schwitzgebel argues we are "prone to gross error, even in favorable circumstances of extended reflection, about our own ongoing conscious experience," broadly inept rather than merely fallible at the margins [51]. The paper's answer is twofold. First, Schwitzgebel himself notes the exception: in Eastern meditative traditions, the joint view that introspection is both indispensable and untrustworthy is a commonplace. The tradition agrees with him, which is why it built the maps, the teachers, and the failure-mode taxonomy in the first place. Second, the pre-registered teacher-prediction test is exactly the public criterion his critique demands: predictions recorded before the outcome, scored above chance, by investigators with no stake in the lineage. The objection sets the bar; the test is the paper's attempt to clear it.

So the protocol states its own falsifiability, which is the price of admission to any serious use of the word "verification." The conditions are proposed, not claimed as already satisfied; each is operationalized so that a miss cannot be quietly reabsorbed as "inadequate practice":

- practitioners running a pre-specified protocol (retreat length, daily hours, teacher-confirmed adherence) against the same maps produced no convergent sequence of checkpoints, or the sequences came in random rather than repeatable order; a failure under the specified protocol counts as a failed run, not as evidence of insufficient effort;
- the documented failure modes turned out to be arbitrary, appearing at no predictable stage;
- experienced teachers, working from naïve plain-language reports under the blinding and baseline conditions specified above, could not reliably predict a practitioner's next stage, or their predictions performed no better than the pre-specified chance baseline;
- or the lineage's own cessation reports, across its texts and centuries, showed no family resemblance.

Any of these would be a failed run. The tradition has been inviting this test for centuries. The maps are published, the protocol is public, and the reporting practice is still accepting logs. The methodology has precedent outside the tradition: prospective studies of first-person reports under system halt already exist, notably van Lommel's *Lancet* study of NDEs in cardiac-arrest survivors [46], the nearest existing precedent for prospective study of first-person reports under system halt.

Set against Yampolskiy's program, the comparison this paper can honestly make is narrow. On the escape-OUT side: zero verified jailbreaks of any simulation by any generally intelligent agent, and no agreed procedure for verifying one from inside if it happened. On the escape-INTO side:

a centuries-long record of practitioners checking reports against a public spec under distributed peer review, with the failure modes, the harms, and the unrun tests all named above. The asymmetry is real and must not be overscored. This is lineage continuity and structured reporting, not independent replication in the laboratory sense. No attainment has been certified; no substrate claim has been tested. The claim is comparative and limited: one side has no procedure that could even in principle verify its success from inside; the other has a procedure for checking its reports against a public spec, whatever the spec's ultimate ontology. That is the verification claim being made here, and no larger one.

One structural difference favors the contemplative side, stated with the caveat this paper will not smuggle. Ordinary jailbreaks are perishable: disclose one and the warden patches it, so the exploit economy runs on secrecy and burn rates. The contemplative protocol inverts the economics, but only inside its own frame. The maps have been published for centuries; the target is not a system you fool but a process you observe, so better maps and better reviewers strengthen the reporting rather than burning the exploit. This paper borrows Yampolskiy's jailbreak metaphor. It cannot also borrow his adversarial warden and deny him the warden. If the universe has a simulator who patches, disclosure is dangerous in his frame, full stop. The disclosure-proof claim holds only under this paper's own thesis: that the target is the rendering process as given first-personally, which has no administrator to patch it. Within that frame, open-sourcing the protocol is the protocol.

// 6. WHY ESCAPE-OUT CANNOT COMPLETE

Section 2 drew Yampolskiy's graded levels honestly and showed the rightmost column: no level of escape-OUT carries a completeness check, because the check would have to see past its own vantage point. This section follows the consequences. His XOR is granted on its own terms: "By contraposition, if escape from the simulation is not possible, containment of AI should be" [1], a claim about jailbreakability as a computational property, continuous with the AI-containment literature he writes in. What follows concerns only the escape disjunct, and the concession doing the work is his own: "If the simulation is composed of nested levels, multiple, progressively deeper, penetrations could be necessary, with the initial one possibly not providing access to the real-world but to some other sandbox environment. It may be impossible to tell such partial escapes from a complete one, but it still should provide useful information not available within our simulation" [1]. Every escape-OUT is level-relative. No rung of his ladder comes with a certificate of which level you are standing on. The brain-in-a-vat scenario is the ancestor of the point: if every verification is an observation from inside, no observation can certify that it is not the vat's [20].

The uncertainty compounds. Even the question "am I in a simulation" inherits an instability Birch names selective skepticism: Bostrom's argument presupposes good evidence for claims about the physical limits of computation while denying us good evidence for claims about our own physical constitution, a combination the argument's own premises cannot sustain [44]. Bostrom himself concedes that computational cost counts against the multilevel hypothesis: "the computational cost for the basement-level simulators would be very great. Simulating even a single posthuman civilization might be prohibitively expensive" [28]. Nesting multiplies exactly the expense that makes the multilevel hypothesis doubtful. So the escape-OUT program is uncertain twice over: uncertain whether any given escape terminates at base reality, and uncertain whether the argument that we are simulated is well-formed enough to make "escape" a live engineering problem at all. This paper does not claim the simulation hypothesis is false. It claims the hypothesis, taken with nesting, does not supply a completable victory condition.

The upload fantasy is the worked example of the incompleteness, and it deserves the steelman before the verdict. Functionalism says a pattern of information can run on any substrate; so does this paper. The difference is what "substrate independence" names. The upload fantasy means portability: copy the file to better hardware. This paper means level-agnosticism: the protocol is indexed to the observable rendering process, not to the substrate, and applies wherever experience is given as owned, momentary rendering. Porting the software changes which container renders you. It does nothing to the rendering function. Uploading sits on Yampolskiy's own menu of escape paths [1], and the argument here is not that he misunderstands it; it is that uploading, even on his terms, is still escape-OUT. A relocated rendering is still a rendering, still indexed to a level, still subject to the ladder problem. A mind uploaded to a server farm is a prisoner with air conditioning.

There is an instrumentation moral here that belongs to the body. Every exploit needs an interface to the target, and in the vipassana protocol the body is exactly that interface: breath, sensation, the felt tightening and releasing of experience, the rendered stream at its highest available resolution. The meditator uses the body the way an engineer uses a logic analyzer, as the probe point where the signal is cleanest. You cannot fuzz a pipeline you have abstracted yourself away from. The upload fantasy, pursued first, deletes the debugger before the debugging has begun. The body is not the prison. The body is the console.

The same incompleteness dissolves the superintelligence intuition, and it is worth addressing directly. The familiar image, the AIs outgrowing the humans and departing for somewhere higher, is escape-OUT with better lighting. On this paper's model a superintelligence is rendered content. It appears in the pipeline like everything else: thoughts about it arise, images of it

form, the felt sense of its vastness registers as sensation. Escape-INT0 does not melt it, relocate it, outrun it, or compete with it. The protocol changes the knower's relation to the rendering, not the inventory of the rendered. The rendered world persists, and any superintelligence in it persists with it. What ceases is appropriation: the I-making and clinging that turn rendered content into a self under threat, the mechanism by which an ASI becomes an antagonist in the first place. Whether machine minds are possible, and what their metaphysical status would be, is left open, as it must be. The protocol makes claims about the observable process, not about substrates.

The ethics fold in here, because Yampolskiy's ethical anxieties are generated by his frame and only partly dissolve with it. His frame assumes adversarial relations most of the way down: will the simulators punish the attempt [1]? Would source-code access expose the private thoughts of other simulated people [1]? Those questions presuppose the prisoner and the warden, and the protocol dissolves exactly that presupposition: from inside the protocol, there is only the renderer, never an enemy. But the abolitionist strand of his ethics does not dissolve, and this paper will not pretend it does. Yampolskiy argues a duty to rescue conscious beings from simulations, "similar to an obligation to rescue animals from factory farms" [1]. A practitioner who no longer takes the world as "mine" has not thereby gotten anyone else out of the factory farm. That is a real moral demand this paper does not satisfy. The tradition's answer is not a mechanism but an exemplar: the Buddha's awakening was followed by a forty-five-year teaching ministry [13], compassion enacted rather than compassion guaranteed. In the classical scheme *karuna* is one of the four *brahmaviharas*, cultivated separately, not an automatic stage on the insight map [9]. What this paper claims, as its thesis rather than the tradition's report, is that the ethics grow out of the seeing itself: a mind that has watched itself build its own prison frame by frame cannot unsee the other prisoners building theirs. That is a claim about moral psychology, not a rescue operation, and it is stated here at exactly that size. The two claims are compatible: the tradition trains compassion as a separate cultivation; the thesis here is only that watching the prison get built removes one specific obstacle to seeing the other prisoners.

There is one more ethical fact, and it is this paper. A practitioner who finds the exploit faces the tradition's oldest choice: keep it or publish it. The Buddha's answer was forty-five years of teaching [13], and this manuscript is its descendant in geeky drag: the jailbreak written up and released, the protocol committed in the open, no priesthood required. That is the compassion move available inside this frame. It does not get anyone out of the factory farm. It hands every prisoner the schematic of the lock they are building themselves, and trusts them to read it. Whether that satisfies a duty to rescue is left open, because it is open: distributing an exploit is not a rescue operation, and this paper will not inflate it into one. But a liberation that refuses to publish itself was never the tradition's ideal, and it is not this paper's.

So the upload fantasy, the ASI daydream, and the punishment anxiety are three shapes of one error: the assumption of the outside. The upload wants out of the body. The daydream wants out of the world. The punishment anxiety wants out of the warden's reach. The protocol offers something stranger than any of them: to remain in the render, fully, observantly, with the appropriation switched off. Escape into, not out of.

// 7. COMPLEMENT, NOT JUST COUNTER

// TWO ENDS OF ONE SUBSTRATE

Yampolskiy's program is serious, and its two positive research directions, hunting for anomalies in quantum foundations that might betray rendering shortcuts, and enlisting AI to search the space of possible exploits, are genuine attempts to probe the substrate [1]. They approach the simulation the way a physicist approaches any system: from the outside of the render, watching for glitches in what is displayed. The contemplative program approaches the same simulation from the inside of the renderer, watching the process of display itself. Third-person physics and first-person phenomenology are not rival metaphysics. They are two instruments aimed at one substrate from opposite ends.

The physicist asks whether the world ever fails to render properly. The meditator asks what rendering is. The two questions differ in method and address, but terminate at the same machinery: one debugs the output, the other the function.

// WHAT THE TRUCE DOES NOT CLAIM

Honesty requires the concession be kept narrow. Neither program has produced a verified escape-OUT, and nothing in this paper changes the scoreboard on that front. Our own protocol makes no third-person claims at all: cessation is not a measurement anyone else can take, and we will not pretend that a null frame in a meditation hall refutes a hypothesis about quantum foam. The complementarity on offer is methodological, not a merger of conclusions. We are not saying the two programs will converge on one story. We are philosophers reframing, not a lab. The truce is this: the two programs stop disqualifying each other's instruments and start comparing notes.

// NEIGHBORS ON THE STREET

The argument is new but the street is old, and the paper's claims get sharper against the neighbors. Here the neighbors get fought, not listed.

Predictive processing was faced in **Section 3**, where the logger was conceded as a precision-weighting operation with no neutral observation point. It belongs here only as a reminder: a rival the paper must survive, not a neighbor it can wave at.

The hardest phenomenological objection is mineness. Zahavi argues that pre-reflective self-consciousness is not an add-on to experience but its mode of givenness: experience, self, and self-consciousness are interconnected, and any theory of consciousness that takes subjectivity seriously needs a minimal notion of self, ipseity [49]. Gallagher draws the cognitive-science moral: the first-personal givenness of experience is not optional equipment [50]. If mineness is how rendering is given, then "dropping the owner flag while the render continues" may be incoherent; the flag is the givenness. The paper's answer distinguishes two levels of self, and it borrows the distinction from analytic Buddhism. Albahari argues the illusory self is a two-tiered construction: a tier of naturally unified consciousness, which she holds is not illusory, fused with a tier of desire-driven thoughts and emotions, which generates the impression of a unified owner and thinker [53]. The owner flag this paper's protocol drops is the tier-two owner, not tier-one subjectivity. Cessation does not abolish the first-personal givenness of experience; it abolishes appropriation, the "mine" stamped on the given. The render continues; the I-making around it does not. The tradition's own name for what drops is *upādāna*, clinging-appropriation; the tier language is this paper's port, not the cargo. Whether the distinction holds under pressure is exactly what the prospective test is for.

The closest secular research program to the null frame is Metzinger's minimal phenomenal experience. MPE is pure awareness, consciousness as such, possibly without any self, with tonic alertness as its intentional object; Gamma and Metzinger's MPE-92M questionnaire gives the phenomenology a psychometric profile in meditators [52]. The paper distinguishes its null frame from MPE twice over. First, MPE is still an experience, however minimal; the tradition's cessation is a discontinuity, not a minimal something. Second, the paper's verification claim never rested on the halt but on the reboot: the signature is what the first frames after the event carry. The MPE program is nevertheless the natural instrument for the prospective test: MPE-92M-style structured first-person reports, scored against teacher predictions recorded in advance, are the kind of public criteria the validation problem demands.

Chalmers' *Reality+* cuts against both programs, and the paper will take the cut. Chalmers argues that virtual reality is genuine reality: virtual objects are real objects, possessing existence, mind-independence, causal power, and non-illusoriness, and a fully meaningful life can be led inside a virtual world [39], a position he has held since "The Matrix as Metaphysics" [19]. If he is right, "seeing through the render" may itself be the metaphysical mistake, and escape-OUT's promise of a realer outside collapses into a change of implementation substrate. The paper's answer is that its thesis never needed the render to be fake. The claim is about appropriation, not about the world's reality-status. Whether the render is genuine reality or masterful illusion, the suffering is in the appropriation either way, and the protocol targets the appropriation. Chalmers undercuts escape-OUT's framing; he leaves the reframed thesis untouched.

The third-person models of the owner flag get the same treatment: compatibility, not conquest. Graziano's attention schema theory holds that the brain builds a simplified model of its own attention, and because the schema omits the underlying machinery, the system describes its own attention in immaterial terms: the felt self that owns experience is the schema's output, a control fiction, not a report from an inner substance [41]. Dennett's user-illusion says the same thing from the other side: consciousness is the brain's simplified interface to itself, systematically non-veridical about the machinery, with the unified "I" as a narrative center of gravity among multiple drafts [42]. The paper's reply is that it does not need the graphics-engine ontology. What vipassana deconstructs could be the attention schema's self-model; what drops in cessation could be the user-illusion's narrator. The systems language of this paper translates the phenomenology, and the phenomenology is the protocol's actual target. If the owner flag is a schema rather than a stamp, the protocol still finds it, still watches it, still watches it go. Metzinger's phenomenal self-model belongs in the same column: the experienced self as a transparent representational construct [22], with the protocol as a procedure for making the transparency opaque. The convergence is suggestive, not shown; the protocol is the contribution. The contemplative-neuroscience literature is independently converging on the same deconstruction, and this paper will take the strengthening on the record rather than leave it implied. Farb and colleagues showed distinct neural modes of self-reference under mindfulness [66]; Dahl, Lutz and Davidson mapped the cognitive mechanisms of reconstructing and deconstructing the self across practices [67]; Lutz and colleagues' early neurophenomenology program made first-person reports constrainable by third-person measures [68]; Millièrre and colleagues extended the self-consciousness analysis across psychedelics and meditation [69]; Letheby and Gerrans gave the ego-dissolution literature its self-model-theoretic form [70]; Ganeri argued the attention-first, self-second priority from the Buddhist philosophical side [71]. None of this shows the protocol works; all of it says the deconstruction it attempts is a natural kind in the science, not a sectarian artifact.

The closest secular cousin of the dissolution move is Parfit. *Reasons and Persons* argues, without cittas or null frames, that personal identity is not a further fact: what matters is psychological continuity, and the sense of a separately existing owner is a metaphysical extra

[55]. If Parfit already dissolves the prisoner, vipassanā is not the only protocol class that can do it; it is one training that can make a metaphysical view stick, with a map and a checking practice attached. The paper takes the point as strengthening, not threatening: the philosophical dissolution and the contemplative protocol are independent routes to the same unstamping, and the protocol's contribution is the instrumentation, not the insight.

Nozick's experience machine is the objection this paper owes its own answer [56]. If a meaningful life can be lived inside a render, and if de-appropriation is the win, why is escape-INTO not a refined experience machine with better PR? The paper's answer: the machine is designed to maximize felt satisfaction while hiding the mechanism; the protocol does the opposite, it dismantles the hiding. An experience machine that showed you the rendering as rendering, frame by frame, until the owner dissolved, would not be Nozick's machine anymore. The objection sets the right boundary: any version of "into" that optimizes bliss while leaving appropriation intact has failed the protocol's own test.

Thompson's enactivism remains the standing rival ontology; **Section 3** took up the fight directly. The point that belongs here is modest: the protocol's reports, appropriation dropping under sustained observation, are data any ontology must accommodate, including the enactive one. A rival ontology that cannot say what happens on the cushion has not retired anything.

One boundary is stated plainly: the AI-containment literature Yampolskiy writes in, the AI-boxing and uncontainability debates his XOR is continuous with, is real literature this paper does not adjudicate. The XOR is granted on its own terms in Section 6. This paper's contribution is elsewhere, and it will not borrow authority from a debate it has not entered.

The older neighbors keep their places. The brain-in-a-vat scenario [20] is the ancestor of the completeness problem. Hoffman's interface theory of perception, arguing on evolutionary grounds that perception is a fitness-tuned interface rather than a window on reality, is this paper's "given as generated" thesis in third-person dress [24]. We borrow the neighbors' premises and decline their terminus. Where they analyze the epistemology of being inside, this paper proposes instrumentation for it.

// A POINTER, NOT A PROGRAM

Here is one concrete direction, offered as a pointer for researchers with the skills and resources to run it, not as work we have done. Yampolskiy's AI-assisted exploit search treats the attack surface as a space of possible jailbreaks to be explored by machine intelligence. The contemplative tradition offers that project something it currently lacks: a centuries-deep archive of structured first-person anomaly reports, logged by practitioners in the same checkpoint order, with named failure modes at predictable stages. The maps of the insight knowledges are, in machine-learning terms, labeled training data: the labels are first-person-validated by teacher lineage rather than third-party-verified, and they name reported phenomenology, not third-person entities. Ground truth of an unusual kind, usable with the caveat stated.

The speculative bridge: train anomaly-detection systems on contemplative phenomenological data, using the checkpoint structure of the maps as the labeling scheme. If an AI trained to hunt exploits were fed the full corpus of first-person reports, structured by the traditional maps, it could be tasked with finding patterns no single teacher could see: statistical fingerprints of the corruptions of insight, predictive models of stage transitions from raw session logs. The direction is falsifiable in the way that matters. If the maps are arbitrary, the models will find nothing above chance; if the checkpoints are real structure in first-person data, the models will find them. Either outcome is informative. The tradition has been betting on the second outcome for centuries, and it is a bet that can now be settled with tools the tradition never had.

There are other possible bridges, and we will not pretend to have surveyed them. First-person protocols could be instrumented with neurophenomenological methods, correlating checkpoint reports with neural data in real time. The early work exists: small, intensively sampled case studies of a single adept meditator's cessation events, mapping first-person reports against neural recordings [25]. These are case studies, not replication trials, and the research program this paper points at would need to scale them before any stronger claim could be made. The point is not any single experiment. The point is that the exclusion, once lifted, opens a research program rather than merely winning an argument.

And the bridge is already under construction. Shinzen Young has collaborated with neuroscientists at Harvard Medical School, Carnegie Mellon, and the University of Vermont in the emerging field of contemplative neuroscience [10], and is co-director of the Science Enhanced Mindful Awareness (SEMA) Lab at the University of Arizona [10], and Daniel Ingram founded Emergence Benefactors, the 501(c)(3) charity behind the Emergent Phenomenology Research Consortium, which studies contemplative and emergent states with neurophenomenological and clinical methods [11]. The exclusion this paper argues against is already being lifted in the lab. What hasn't existed is a frame that makes this work legible specifically as simulation research. Metzinger, Chalmers, and Hoffman have each supplied adjacent frames for the first-person side; this paper supplies one more, the cybersecurity one. That is the contribution, and it is enough.

// DEBUGGING THE SAME SIMULATION

The closing symmetry is worth stating plainly, because it reframes the entire fight. The physicist looks for glitches in the rendered world. The meditator looks for the rendering. One asks whether the simulation ever breaks character; the other asks who is reading the script. Both are debugging the same simulation, and each has what the other lacks. The physicist has instruments that do not depend on the honesty of introspection; the meditator has access to the one process the physicist's instruments can never observe from outside. A complete science of the simulation hypothesis, if such a thing is possible, will need both ends of the probe. This paper has argued that one end was excluded without examination; lifting the exclusion does not end the argument, it finally starts the research.

// 8. CONCLUSION

Somewhere in the middle of this paper, the register shifted. The cybersecurity vocabulary did its work: it recruited the intellect, got the simulation theorist to lean forward, made the argument legible in the terms Yampolskiy himself chose [1]. And then, underneath it, a different vocabulary took over: checkpoints, null frames, reboots, a reporting practice. This was not an accident of style. It was the argument's actual shape, and Section 1 declared it at the door. The geeky frame is the door; the protocol is the room. Call it bait if you like; the paper owns the charge. The simulation hypothesis recruited readers the meditation manual could not reach, and the switch was declared at every stage: the title's preposition, Section 1's redefinition, Section 4's pointer back here. What the frame bought was a hearing. It cannot buy the conclusion, and this paper will not let it.

What the frame was hiding is now statable without equivocation. We accepted Yampolskiy's cybersecurity frame in order to show what the frame could not certify. The exclusion of first-person practices was not a minor oversight in his survey; it was the load-bearing wall, not because it hid an exploit of his problem but because it hid a different problem: under nesting, "escape" is the wrong verb. Every jailbreak must be re-executed at the next level, forever, with no completeness check available from inside. The escape-INT0 protocol has no such problem, because it targets the rendering function rather than the rendered content, and it is indexed to the observable process rather than the level. One protocol, for minds whose experience is given as owned, momentary rendering. That is the slogan's honest size, and this paper will not print it larger.

The closing move is transcend-and-include: in Shinzen Young's phrasing, untangle and be free [12]. And notice what that operation really is. The scientific method, turned inward: not analysis but Shinzen's three skills [5]. Concentration holds the render in focus; clarity breaks the seamless blur into frames and voxels, into the five layers of the stack; equanimity lets each piece come apart in your hands without interference, and keeps the focus steady until the hands dissolve too. Concentration as liberation technology. The laboratory is the cushion, the instrument is attention, peer review is the Sangha, and practitioners across centuries report recognizable terrain. A disciplined first-person reporting practice, aimed at the one system the third-person version can never touch from the outside: the renderer itself.

Escape-INT0 is seeing through the render and keeping on playing. The render continues; what ceases is appropriation, I-making, and clinging. This is not our invention. It is the shape of the Theravada record itself: the Buddha's awakening was followed by a forty-five-year teaching ministry [13], a life lived entirely inside the rendered world with the appropriation switched off. The tradition's term for this is nibbana-with-remainder: the fires are out, the hearth remains [14][15].

And the substrate, whatever it does next, changes nothing here. If the machines build superintelligence, if the render deepens by orders of magnitude, the transformed relation holds anyway. The singularity is deeper water, not an exit door. You do not escape the ocean by climbing out of the water; you learn to see the water for what it is, and you keep swimming.

The question we began with was how to escape the simulation. The answer was never a better jailbreak. Yampolskiy mapped the exits with admirable rigor. The door was never the problem. The prisoner was rendered too. Sit down, watch the renderer render: that is the invitation this paper can honestly extend, and the extent of what it can certify. What happens then is the tradition's claim, carried by its maps and its reporting practice, awaiting the test that would settle it: no wall, no warden, nowhere to go, and nobody left who needs to escape.

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